

Spin Peierls Transition of $J_1 - J_2$ and Extended Models with Ferromagnetic J_1

Manodip Routh¹, Sudip Kumar Saha¹, Manoranjan Kumar¹ & Zoltán G. Soos²

Department of Condensed Matter Physics and Material Sciences

¹S. N. Bose National Centre for Basic Sciences, India,

²Princeton University, USA

Introduction

- Spin-Peierls (SP) transition, where the lattice dimerizes due to its coupling to the spin systems, is the particular quantum mechanical nature of quasi one-dimensional (1D) antiferromagnets.
- We discuss SP transition of the ferromagnetic $J_1 - J_2$ model, which differs from the antiferromagnetic case.
- We obtain the thermodynamic limit of the spin susceptibility $\chi(T)$ and specific heat $C(T)$ of zigzag chains in β -TeVO₄, a recently identified quasi-1D material

Electronic Instability

- The Hamiltonian of the chain dimerized $J_1 - J_2$ model is,

$$H(\alpha, \delta) = \sum_r -(1 - \delta(-1)^r) \vec{S}_r \cdot \vec{S}_{r+1} + \alpha \vec{S}_r \cdot \vec{S}_{r+2}$$

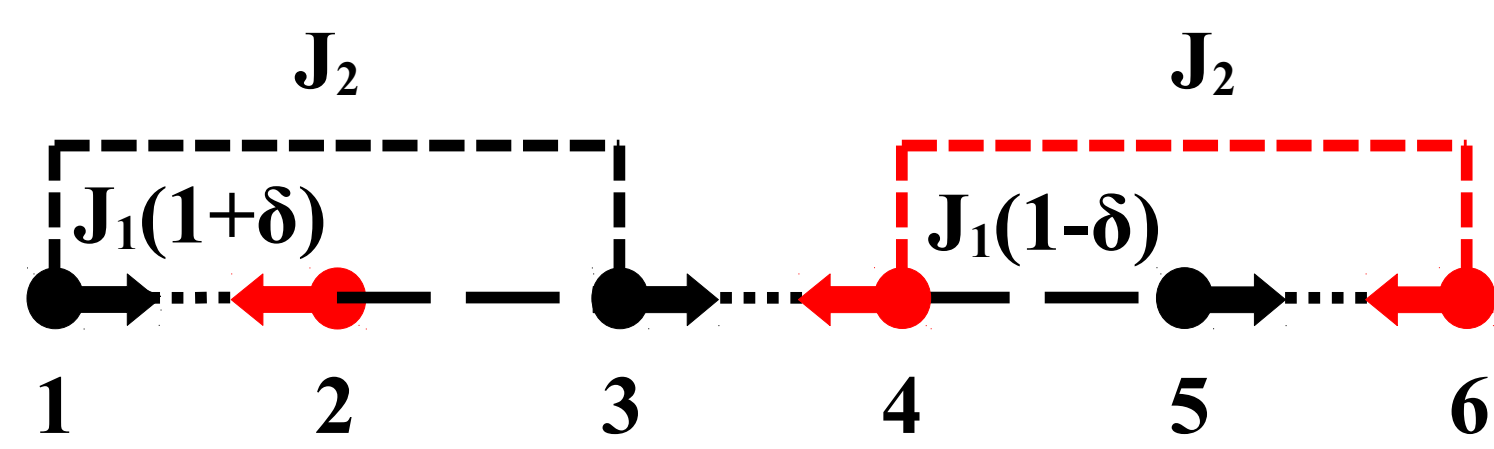


Figure 1: Dimerized chain with alternating $J_1(1 \pm \delta)$ and unchanged J_2 .

- The Hamiltonian of the sublattice dimerized extended ($\gamma > 0$) $J_1 - J_2$ model is,

$$H_S(\alpha, \gamma, \delta) = \sum_r - \left(1 - \gamma(-1)^r - \delta \cos \frac{\pi r}{2}\right) \vec{S}_r \cdot \vec{S}_{r+1} + \alpha \left(1 + \sqrt{2} \delta \sin \frac{\pi(2r-1)}{4}\right) \vec{S}_r \cdot \vec{S}_{r+2}$$

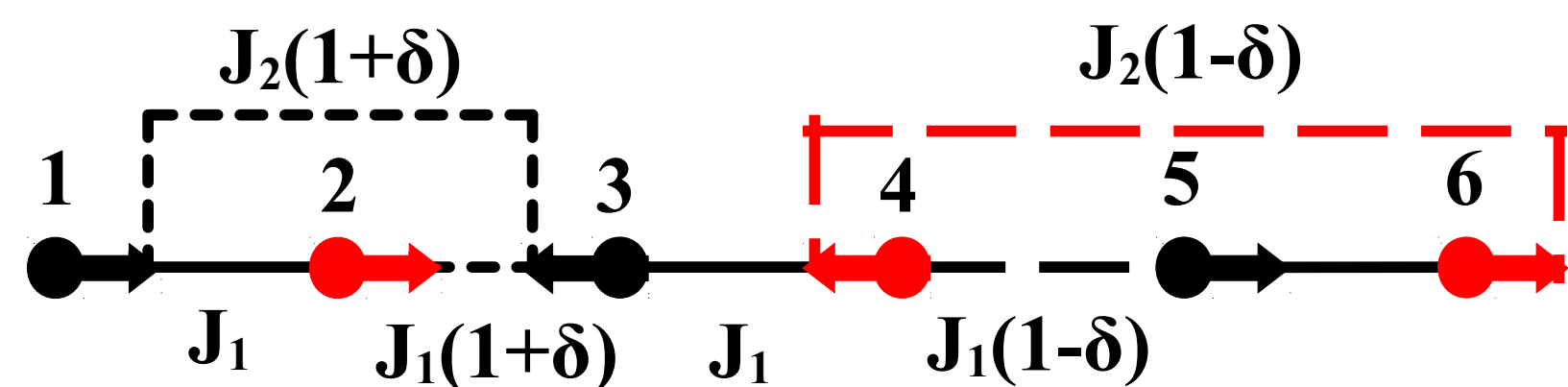


Figure 2: Sublattice dimerization with alternating $J_2(1 \pm \delta)$.

- The instability of the ferromagnetic $J_1 - J_2$ model in the singlet sector with $\alpha > \alpha_c = 1/4$ is to sublattice dimerization at large α when exchange between second neighbors dominates and to chain dimerization at intermediate α .

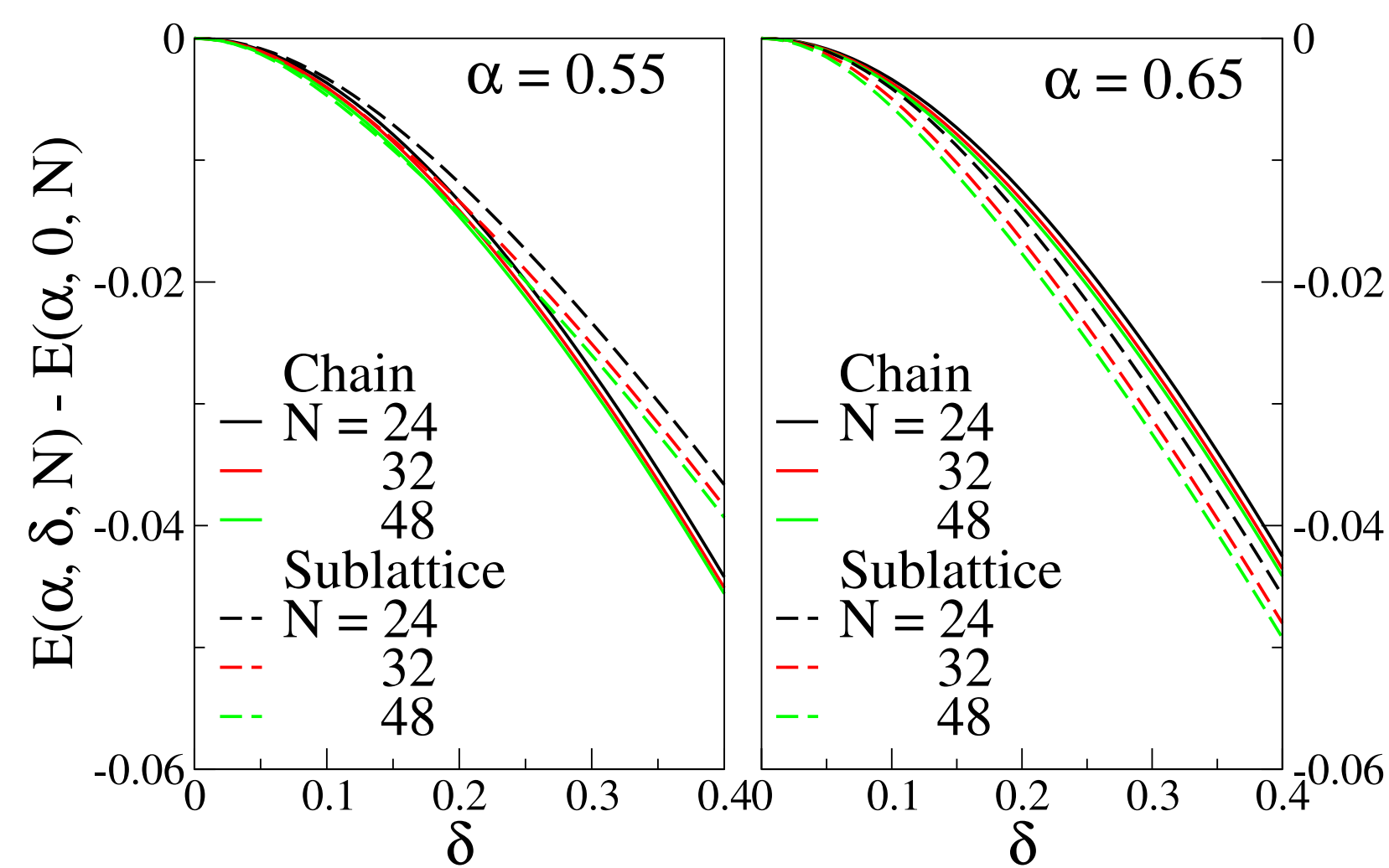


Figure 3: Ground state stabilization per site of the ferromagnetic $J_1 - J_2$ model for $\alpha = 0.55$ or 0.65 . Solid lines and dashed lines are for chain dimerization and sublattice dimerization.

Thermodynamics

- The thermodynamical quantities are obtained using **Hybrid ED/DMRG** technique. Converged thermodynamics are accessible for $T > 0.03$ at system sizes up to $N \sim 100$.

Entropy Convergence

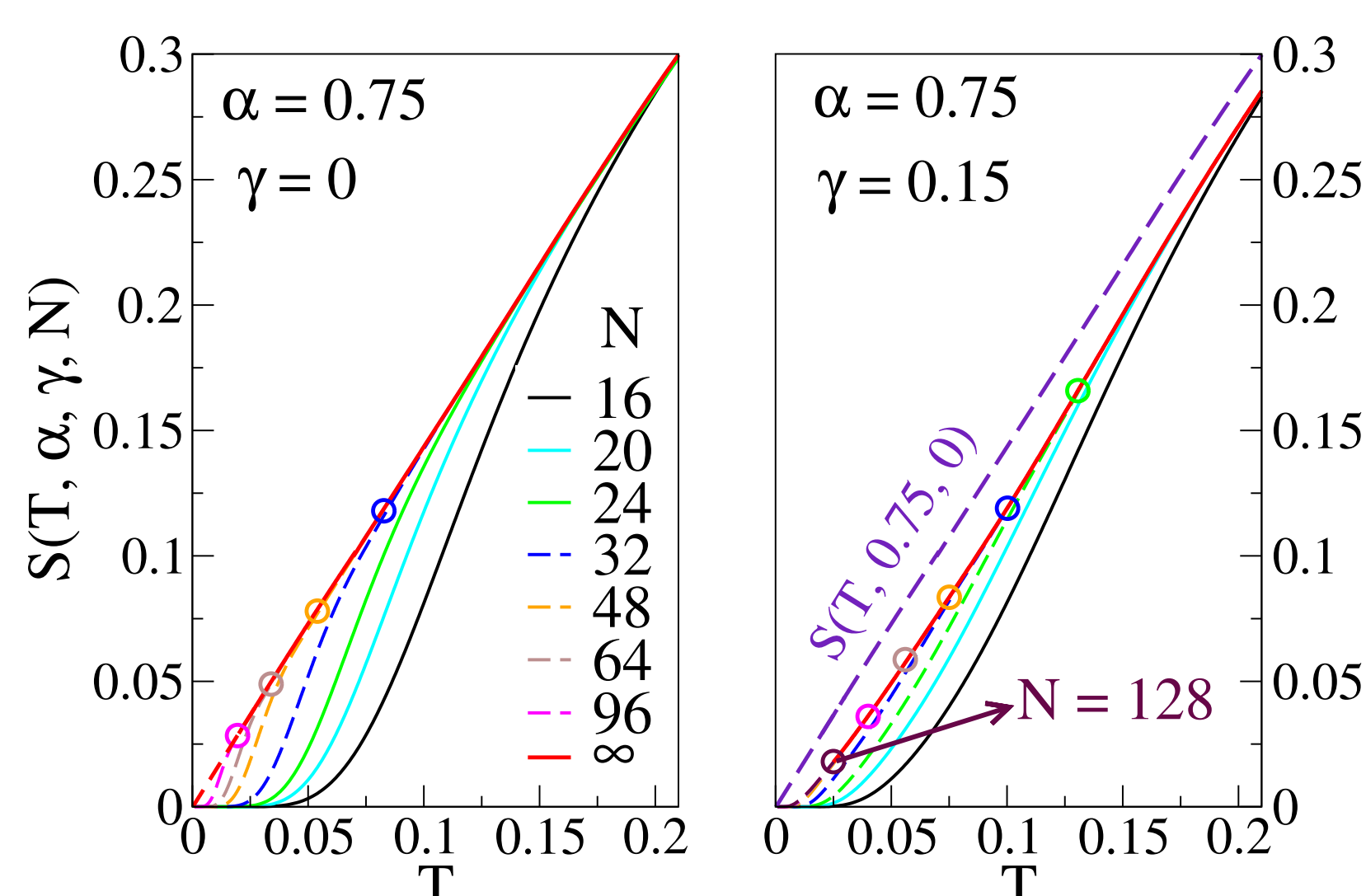


Figure 4: Solid lines are ED upto $N = 24$ for $\gamma = 0$ and 20 for $\gamma = 0.15$.

Susceptibility Convergence

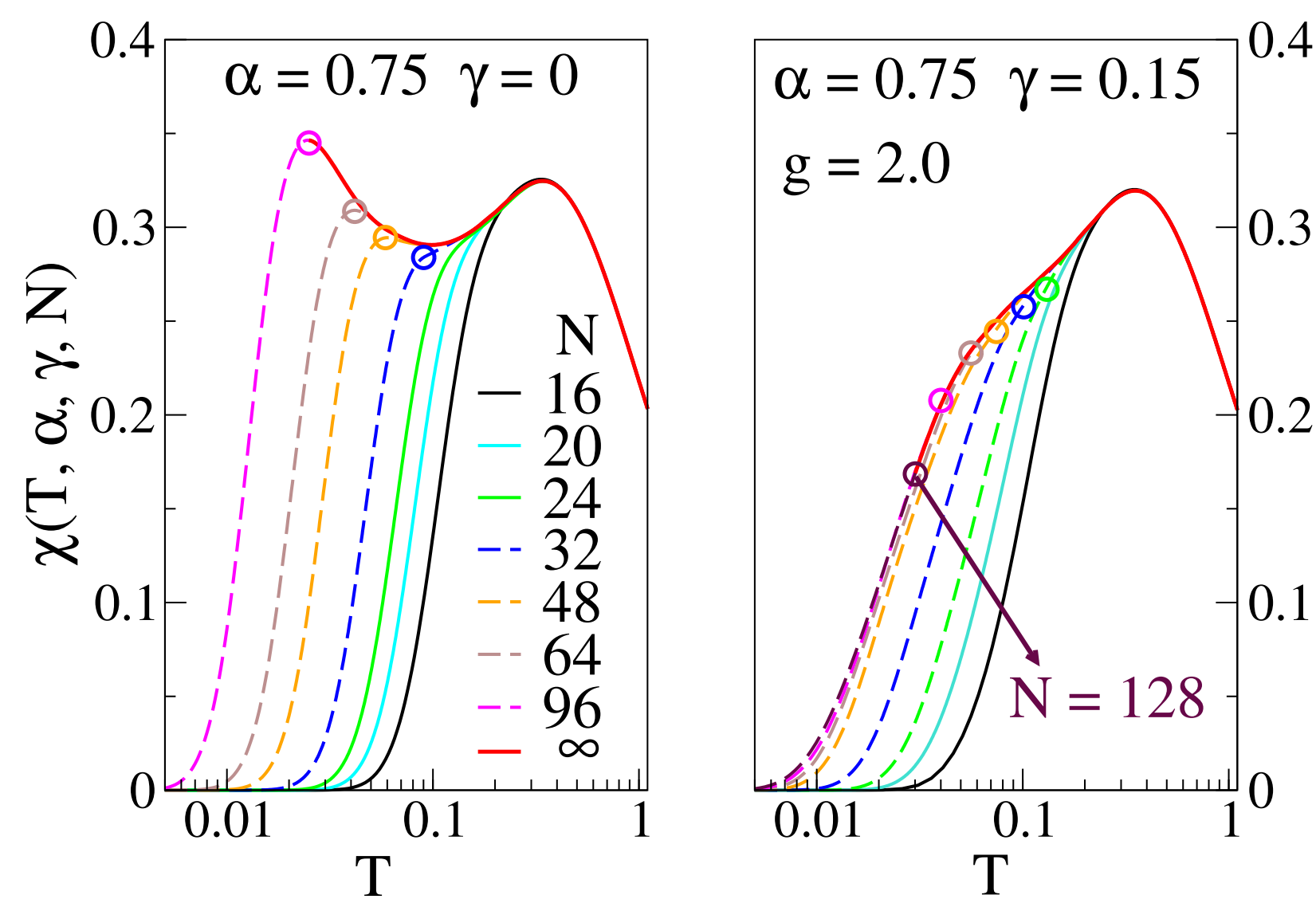


Figure 5: Solid lines are ED upto $N = 24$ for $\gamma = 0$ and 20 for $\gamma = 0.15$.

- Entropy $S(T, \alpha, \gamma, N)$ and magnetic susceptibility $\chi(T, \alpha, \gamma, N)$ at $\alpha = 0.75$ with $\gamma = 0$ and 0.15 . The $\gamma = 0.15$ chain is clearly gapped.

Spin-Peierls Transition

Equilibrium Dimerization

- The equilibrium sublattice dimerization for $\alpha > 0.65$ is

$$\frac{\delta(T)}{\epsilon_d} = - \left(\frac{\partial A(T, \delta)}{\partial \delta} \right)_{\delta(T)} \quad T \leq T_{SP}$$

$A(T, \delta)$ is the free energy per site of the system under consideration.

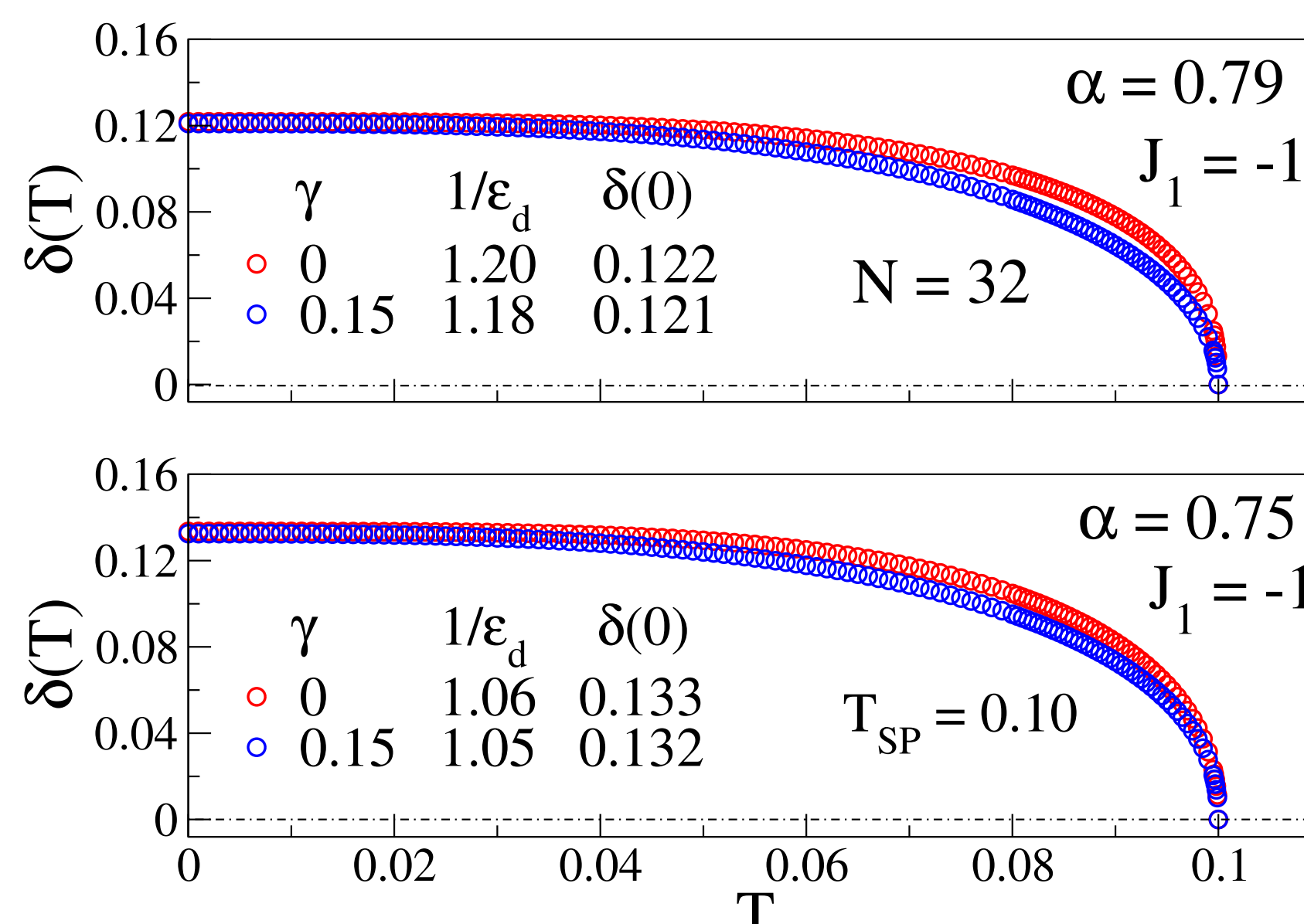


Figure 6: $\delta(T)$ of the converged sublattice dimerization in spin chains with $T_{SP} = 0.10$.

Specific Heat in SP Transition

- The anomaly in $C(T)$ for $T \leq T_{SP}$ is evaluated as

$$\frac{C(T)}{T} \equiv \frac{dS(T, \delta(T))}{dT} = \frac{\partial S(T, \delta(T))}{\partial T} + \frac{d\delta(T)}{dT} \frac{\partial S(T, \delta(T))}{\partial \delta}$$

- The first term is $C(T, \delta(T))/T$ at the equilibrium $\delta(T)$. The second term is discontinuous at T_{SP} where $\delta(T) = 0$ for $T \geq T_{SP}$.

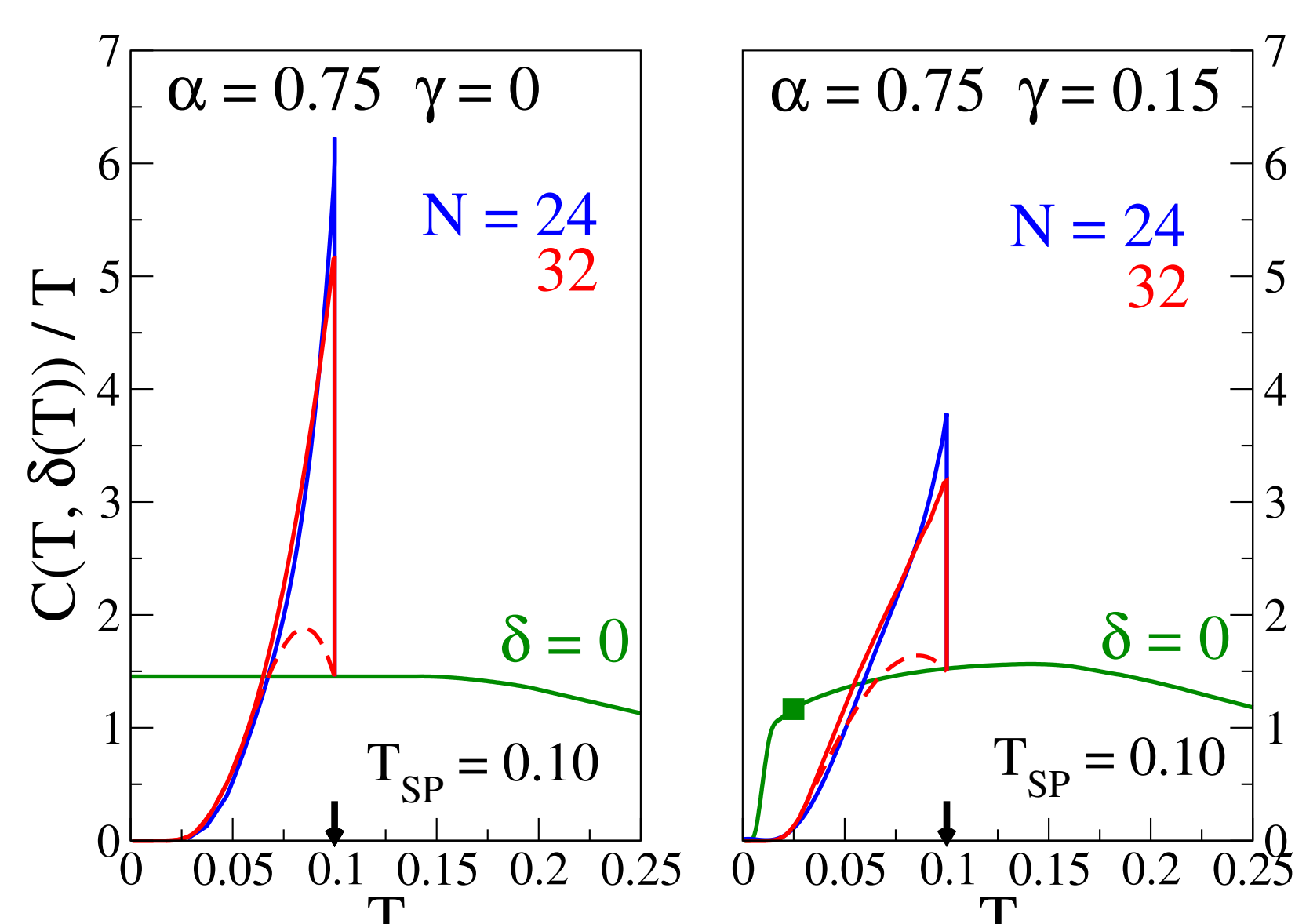


Figure 7: Specific heat, $C(T, \alpha, \gamma, N)/T$ with $\alpha = 0.75$, for the $J_1 - J_2$ model ($\gamma = 0$) and the extended ($\gamma = 0.15$) model.

- The smaller $C(T)/T$ anomaly for $\gamma = 0.15$ is primarily due to the singlet-triplet gap $\Delta(\alpha, \gamma)$.

Susceptibility in SP Transition

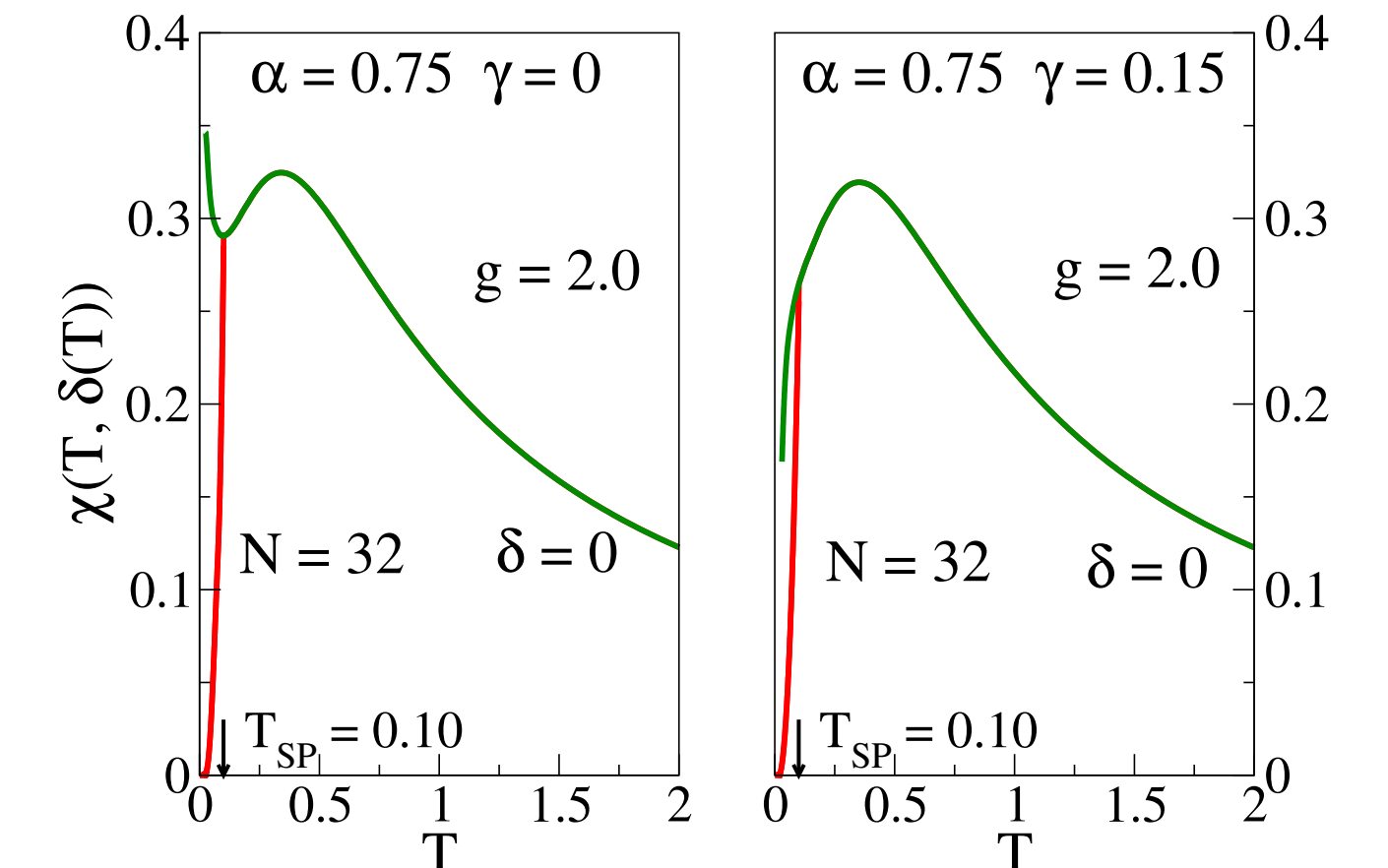
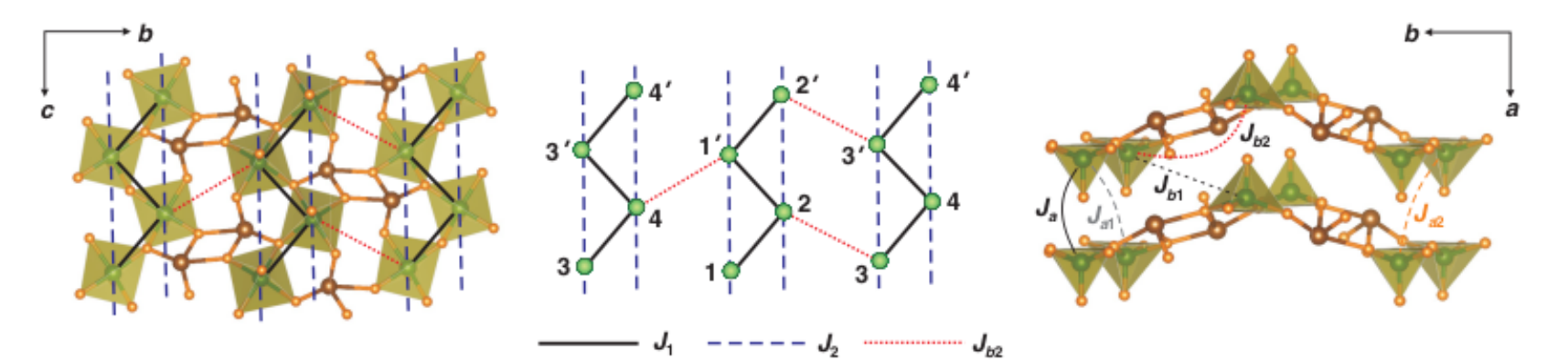


Figure 8: Magnetic susceptibility, $\chi(T, \alpha, \gamma, N)$ with $\alpha = 0.75$, for the $J_1 - J_2$ model ($\gamma = 0$) and the extended ($\gamma = 0.15$) model.

- The SP transition generates a discontinuous $\chi(T)$ slope at T_{SP} . The cusp of the $\gamma = 0.15$ chain is reduced since sublattice dimerization merely increases the $\delta = 0$ gap.

β -TeVO₄ Thermodynamics



- β -TeVO₄ is a quasi 1D quantum magnet. The zigzag chain formed by the distorted VO₅ pyramids sharing their corners in bc plane.

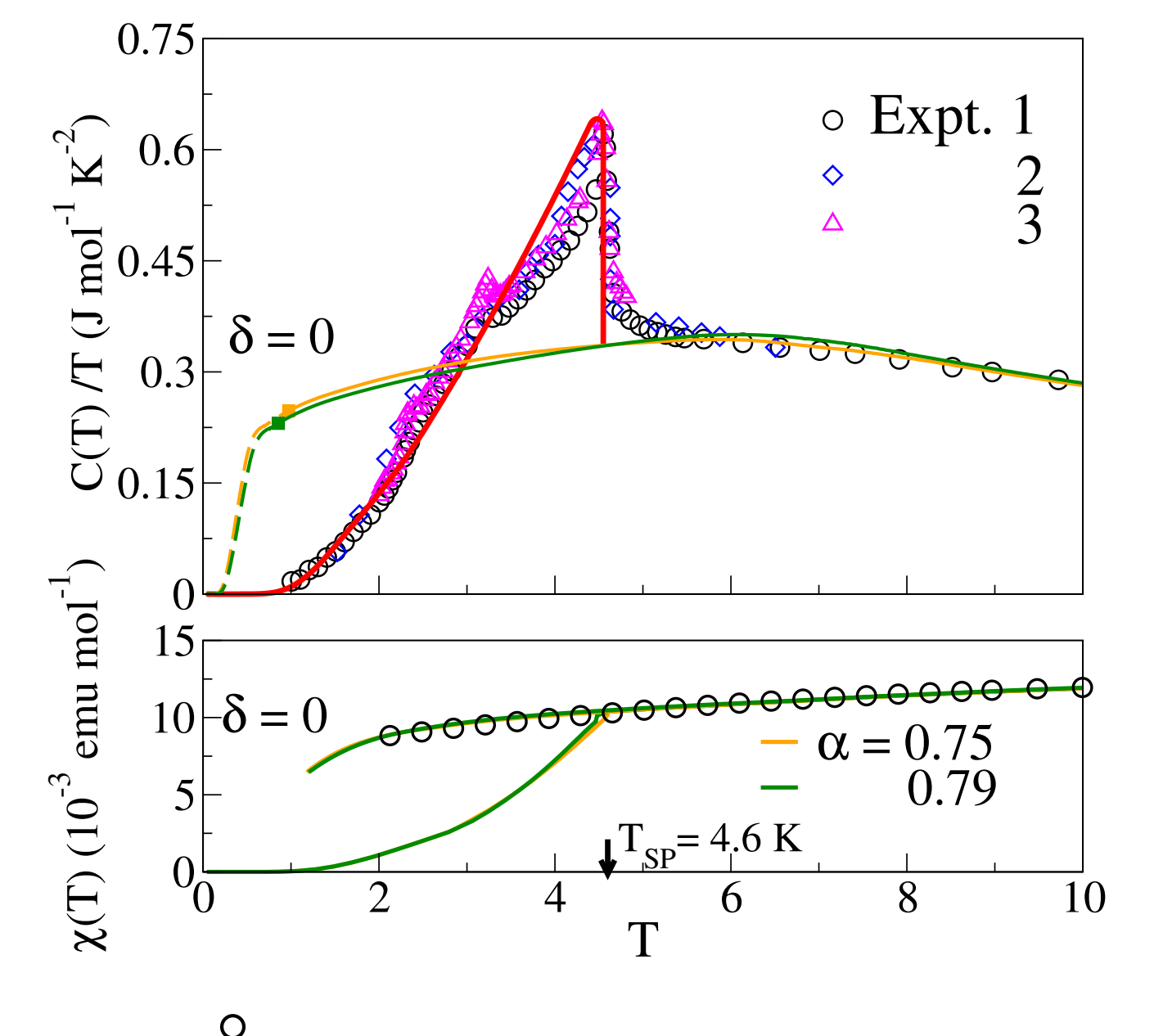


Figure 9: Upper panel: Measured $C(T)/T$ for $T < 6$ K and $T < 10$ K. Lower panel: Measured $\chi(T)$.

- The thermodynamics of β -TeVO₄ has been done previously based on the $J_1 - J_2$ model and ED for $N < 20$ spins with different J_1 and $\alpha \sim 0.77 - 1.0$. Finite size effect appear in $\chi(T)$ at $T \sim 0.15$ or $T \sim 6$ K for $-J_1 \sim 40$ K.
- A remarkable feature of the 4.6 K transition is a clear $C(T)/T$ anomaly with hardly any $\chi(T)$ signature.
- The fundamental issue is that the $C(T)/T$ data require a larger gap at low T than in $\delta = 0$ models that yield the thermodynamics at higher T .
- An increased gap necessarily reduces $\chi(T)$. The reconciliation of a small $\chi(T)$ gap and large $C(T)$ gap is a serious problem for quantitative modeling of β -TeVO₄.

Conclusions

- The spin-Peierls instability of the ferromagnetic $J_1 - J_2$ model in the singlet sector with $J_1 < 0$ and $\delta = 0$ differs in several ways from SP systems with $J_1 > 0$.
- The extended model with $\gamma > 0$ has a finite singlet-triplet gap $\Delta(\alpha, \gamma)$. The SP instability is conditional and the transition with $\alpha > 0.65$ is to sublattice dimerization.
- The large gap due to sublattice dimerization is incompatible with $\chi(T)$. The $\delta = 0$ curves in the lower panel account for $\chi(T)$ down to 2 K without an SP transition at 4.6 K. The same J_1, α, γ and $T_{SP} = 4.6$ K generate the red $\gamma = 0.15$ curves in the upper panel.

References

- [1] Manodip Routh, Sudip Kumar Saha, Manoranjan Kumar, and Zoltán G. Soos. Spin-peierls transition of $J_1 - J_2$ and extended models with ferromagnetic J_1 : Sublattice dimerization and thermodynamics of zigzag chains in β -tevo₄. *Phys. Rev. B*, 105:235109, Jun 2022.